Pytest Exercise — FastAPI API Testing

Task Manager API.

Create:

```
pytest_fastapi_assignment
```

Create:

```
main.py
test_main.py
```

Install:

```
python -m pip install fastapi pytest httpx
```

In `main.py` , build a simple FastAPI application with a list of tasks.

Use fields:

```
id
title
status
```

Sample records can include:

```
1 | Prepare report | Pending
2 | Review invoice | Completed
3 | Update dashboard | Pending
```

Your API should support:

```
GET    /tasks
GET    /tasks/{task_id}
POST   /tasks
```

```
PUT    /tasks/{task_id}
DELETE /tasks/{task_id}
```

Pytest assignments

1. Create a `TestClient` for the FastAPI application.

2. Write a test for:

```
GET /tasks
```

Check that the status code is `200` .

3. Check that the response from `GET /tasks` is a list.

4. Write a test for:

```
GET /tasks/1
```

Check:

- status code is `200`
- returned `id` is `1`

5. Test a task ID that does not exist.

Example:

```
GET /tasks/999
```

Check that the API returns:

```
404
```

6. Write a test for:

```
POST /tasks
```

Use:

```
{
  "id": 4,
  "title": "Check server logs",
  "status": "Pending"
}
```

Check:

- status code
- returned ID
- returned title

7. Send invalid POST data by omitting one required field.

Check that FastAPI returns a validation error.

8. Write a test for:

```
PUT /tasks/2
```

Change the status to:

```
Completed
```

Verify the response.

9. Write a test for:

```
DELETE /tasks/3
```

Verify that the response indicates successful deletion.

10. After deleting task `3`, call:

```
GET /tasks/3
```

and verify that it returns `404`.

11. Run all tests using:

```
python -m pytest -v
```

12. Intentionally modify one endpoint so that one expected value changes.

Run pytest again and identify:

- which test failed
- which assertion failed
- expected value
- actual value

13. Fix the API code and rerun the tests.

All tests should pass again.

The main learning flow is:

```
Write API
   ↓
Write tests
   ↓
Run pytest
   ↓
Change API code
   ↓
Test fails
   ↓
Fix code
   ↓
Test passes
```